

Recall

Linear Transformation :- Let V and W be two vector spaces over F .
A map $T: V \rightarrow W$ is called linear transformation if

① $T(v_1 + v_2) = T(v_1) + T(v_2)$ ② $T(\alpha v_1) = \alpha T(v_1) \quad \forall v_1, v_2 \in V, \alpha \in F$.

Aim :- Properties of linear transformations :-

① $\text{Range}(T) := \{ T(v) \mid v \in V \} \subseteq W$

② $\text{Nullspace}(T) := \{ v \in V \mid T(v) = 0 \} \subseteq V$.

* $\text{Range}(T)$ is a subspace of W .

(i) $\text{Range}(T) \neq \emptyset \quad ; \quad 0 \in \text{Range}(T)$

• T.I.P $T(0) = 0$

$T(0) = T(0+0) = T(0) + T(0)$

$T(0) = T(0) + T(0) \Rightarrow \underline{T(0) = 0}$

(ii) If $w_1, w_2 \in \text{Range}(T)$, then $w_1 + w_2 \in \text{Range}(T)$

$\because w_1 \in \text{Range}(T) \Rightarrow \exists v_1 \in V \text{ s.t. } T(v_1) = w_1$

$\because w_2 \in \text{Range}(T) \Rightarrow \exists v_2 \in V \text{ s.t. } T(v_2) = w_2$

$T(v_1 + v_2) = T(v_1) + T(v_2) = w_1 + w_2$
 $\Rightarrow w_1 + w_2 \in \text{Range}(T)$.

(iii) If $w_1 \in \text{Range}(T)$, then $\alpha w_1 \in \text{Range}(T)$
 $T(\alpha v_1) = \alpha T(v_1) = \alpha w_1 \Rightarrow \alpha w_1 \in \text{Range}(T)$

* $\text{Range}(T)$ is a subspace of W .

* $N(T)$:= Nullspace (T) is a subspace of V .

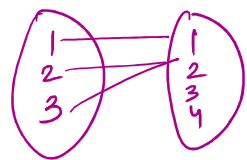
(i) $0 \in N(T)$

$\because T(0) = 0 \Rightarrow N(T) \neq \emptyset$.

$\because v_1, v_2 \in N(T) \Rightarrow T(v_1) = 0 \quad \& \quad T(v_2) = 0$

$f: A \rightarrow B$

$\text{Dom} f = \{1, 2\}$



$\text{Im} f = \text{Range}(T)$

$$(i) \quad 0 \in N(T)$$

$$(ii) \quad \text{let } v_1, v_2 \in N(T) \Rightarrow T(v_1) = 0 \text{ and } T(v_2) = 0$$

$$T(v_1 + v_2) = T(v_1) + T(v_2) = 0 + 0 = 0$$

$$(iii) \quad \text{let } v_1 \in N(T), \quad \underline{T.v.} \quad \alpha v_1 \in N(T) \quad \forall \alpha \in F.$$

$$\because v_1 \in N(T) \Rightarrow T(v_1) = 0$$

$$T(\alpha v_1) = \alpha T(v_1) = \alpha \cdot 0 = 0 \Rightarrow \alpha v_1 \in N(T)$$

• $N(T)$ is a subspace of V .

* Dimension of $\text{Range}(T)$ is called rank of linear transformation T .

* Dimension of Nullspace(T) is called Nullity of linear transformation T .

Example :- $T: \mathbb{R}^3 \rightarrow \mathbb{R}^3$ defined by

$$T(x, y, z) = (y, z, y - z)$$

• Check T is a linear transformation

$$(i) \quad T((x_1, y_1, z_1) + (x_2, y_2, z_2)) = T(x_1, y_1, z_1) + T(x_2, y_2, z_2)$$

$$(ii) \quad T(\alpha(x_1, y_1, z_1)) = \alpha T(x_1, y_1, z_1)$$

$$\text{Range}(T) := \{ T(x, y, z) \mid (x, y, z) \in \mathbb{R}^3 \}$$

$$\text{Range}(T) = \{ (y, z, y - z) \mid y, z \in \mathbb{R} \}$$

Basis for $\text{Range}(T)$:-

• If $T: V \rightarrow W$ is linear transformation. Choose a basis for V . Let $\{v_1, v_2, \dots, v_n\}$ be a basis of V .
 $\therefore \{T(v_1), T(v_2), \dots, T(v_n)\}$ contains a basis of

a basis for V then $\{T(v_1), T(v_2), \dots, T(v_n)\}$ contains a basis of $\text{Range}(T)$.

$$T: \mathbb{R}^3 \rightarrow \mathbb{R}^3$$

$$T(x, y, z) = (y, z, y-z)$$

$$T(1, 0, 0) = \cancel{(0, 0, 0)} \quad \text{NOT L.I. set}$$

$$T(0, 1, 0) = (1, 0, 1)$$

$$T(0, 0, 1) = (0, 1, -1)$$

• show that $\{(1, 0, 1), (0, 1, -1)\}$ is basis of $\text{Range}(T)$.

L.I. set $\left\{ \begin{aligned} \alpha(1, 0, 1) + \beta(0, 1, -1) &= (0, 0, 0) \\ (\alpha, \beta, \alpha - \beta) &= (0, 0, 0) \end{aligned} \right. \Rightarrow \boxed{\alpha = 0, \beta = 0} \quad \alpha - \beta = 0$

spanning set $\left\{ \begin{aligned} (y, z, y-z) &= y(1, 0, 1) + z(0, 1, -1) \end{aligned} \right.$

* $\{(1, 0, 1), (0, 1, -1)\}$ is a basis of $\text{Range}(T)$
 $\text{Rank}(T) = 2$.

Nullity of T

$$\text{Nullspace}(T) := \left\{ (x, y, z) \in \mathbb{R}^3 \mid T(x, y, z) = (0, 0, 0) \right\}$$

$$= \left\{ (x, y, z) \in \mathbb{R}^3 \mid (y, z, y-z) = (0, 0, 0) \right\}$$

$$\quad \quad \quad \Downarrow$$

$$\quad \quad \quad y=0, \quad z=0$$

$$= \left\{ (x, 0, 0) \mid x \in \mathbb{R} \right\}$$

Basis $\longrightarrow \{(1, 0, 0)\}$ is basis of $N(T)$.

Basis $\rightarrow \{(1,0,0)\}$ is basis of $N(T)$.

• $\alpha(1,0,0) = (0,0,0) \Rightarrow \alpha=0$

• $(x,0,0) = x(1,0,0)$

$\text{Nullity}(T) = 1$

• $T: \mathbb{R}^3 \rightarrow \mathbb{R}^3$

$\text{Dim}(\mathbb{R}^3) = 3$

$\text{Dim}(\text{Range}(T)) = 2 = \text{Rank}(T)$

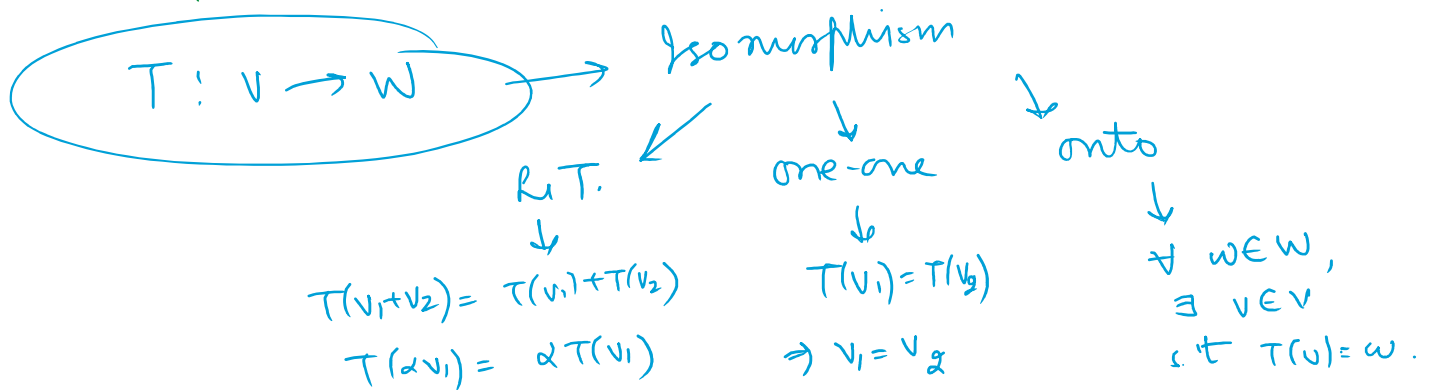
$\text{Dim}(N(T)) = 1 = \text{Nullity}(T)$

$\text{Dim}(\mathbb{R}^3) = \text{Rank}(T) + \text{Nullity}(T)$

• Rank - Nullity Theorem:- Let $T: V \rightarrow W$ be a linear transformation. Then

$\text{Dim}(V) = \text{Rank}(T) + \text{Nullity}(T)$

Isomorphism:- Let V and W be two vector spaces over F . A bijjective linear transformation $T: V \rightarrow W$ is called Isomorphism.



- If $\exists$ an isomorphism b/w vector spaces V and W , then V and W are isomorphic vector spaces.
- If V and W are isomorphic v.s then $\text{dim}(V) = \text{dim}(W)$.

• $T: \mathbb{R}^3 \rightarrow \mathbb{R}^3 \quad T(x, y, z) = (y, z, y-z)$

Is it isomorphism?? **(No)**

Not one-one

$$T(1, y, z) = (y, z, y-z)$$

$$T(2, y, z) = (y, z, y-z)$$

$$T(1, y, z) = T(2, y, z) \quad \text{but} \quad (1, y, z) \neq (2, y, z)$$

Not onto $\rightarrow (1, 2, 0) \in \mathbb{R}^3, \quad (1, 2, 0) \notin \text{Range}(T)$

• Example of Isomorphism

$$T: \mathbb{R}^4 \rightarrow M_2(\mathbb{R})$$

$$T(a, b, c, d) = \begin{bmatrix} a & b \\ c & d \end{bmatrix}$$

L.T

* Check it is an isomorphism.

• First Isomorphism Thm :-

Let $T: V \rightarrow W$ be a linear transformation. Let $N(T)$ denotes Null space of T . Then the quotient space $\frac{V}{N(T)}$ is isomorphic to $\text{Range}(T)$.

$N(T)$ is a subspace V .

• Rank Nullity Theorem

T.I.P $\dim(V) = \text{Rank}(T) + \text{Nullity}(T)$

• $\dim(V/W) = \dim V - \dim W$

• If V and W are isomorphic v.s. then $\dim(V) = \dim W$.

- Dim(V)
- First Isomorphism Thm

then $\dim(v) = \dim w$.

$$\Rightarrow \dim\left(\frac{V}{N(T)}\right) = \dim(\text{Range}(T))$$

$$\Rightarrow \dim(V) - \dim(N(T)) = \dim(\text{Range}(T))$$

$$\Rightarrow \dim(V) = \text{Rank}(T) + \text{Nullity}(T).$$

Sketch of second proof of Rank-Nullity Thm

$$\dim(V) = \underbrace{\text{Nullity}(T)}_{\dim(N(T))} + \underbrace{\text{Rank}(T)}_{\dim(\text{Range}(T))}$$

$N(T)$ is a subspace of V .

Choose a basis $\{v_1, v_2, \dots, v_k\}$ of $N(T)$

$\{v_1, v_2, \dots, v_k\}$ is a L.I. subset of V .

Any L.I. set can be extended to basis

let $\{v_1, v_2, \dots, v_k, \underbrace{v_{k+1}, v_{k+2}, \dots, v_n}_{\text{basis of Range}(T)}\}$ is basis of V .

Claim $\{T(v_{k+1}), T(v_{k+2}), \dots, T(v_n)\}$ is basis of $\text{Range}(T)$.

$$T(x, y, z) = (y, z, y-z) \quad N(T) = \{(x, 0, 0) \mid x \in \mathbb{R}\}$$

$$\text{Basis of } N(T) = \{(1, 0, 0)\}$$

$$\bullet \{(0, 0, 0)\} \text{ is L.I. set in } \mathbb{R}^3$$

$$\bullet \{(1, 0, 0), (0, 1, 0), (0, 0, 1)\} \text{ is a basis of } \mathbb{R}^3$$

$\bullet \{ \underline{(1,0,0)}, \underline{(0,1,0)}, \underline{(0,0,1)} \}$ is a basis of $\text{Range}(T)$
Claim $\{ T(0,1,0), T(0,0,1) \}$ is a basis of $\text{Range}(T)$
 $\quad \quad \quad \parallel \quad \quad \quad \parallel$
 $\quad \quad \quad (1,0,1) \quad \quad \quad (0,1,-1)$ (we have seen this already).